

Aditya Shirsatrao

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Solapur, Maharashtra, India

Education

N. K. Orchid College of Engineering & Technology Solapur, Maharashtra
Bachelor of Technology in Artificial Intelligence and Data Science – CGPA: 8.5/10 Aug. 2023 – Present (Expected 2027)
Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Distributed Systems, Machine Learning, Neural Networks, OOP.

Experience

Fullstack Web Development Intern Unified Mentor Pvt. Ltd
Backend, Distributed Systems & AI-Integrated Products Jul. 2025 – Jan. 2026
• Architected an AI-powered code review platform processing GitHub pull requests asynchronously; integrated Kafka message queues and Redis caching, reducing analysis latency by 45% through parallel distributed workers and AST-based code search.
• Engineered a real-time collaborative IDE with sub-100ms sync using WebSockets, Socket.IO, Redis pub/sub, and CRDT-based operational transforms.
• Built a distributed ML inference platform with autoscaling endpoints, model versioning, and Prometheus monitoring; improved throughput by 60% under concurrent load via request batching and inference caching.
AI/ML Lead Google Developer Groups on Campus (GDG-NKOCET)
Community Leadership & Mentorship Sep. 2025 – Present
• Lead AI/ML initiatives and technical workshops for a 200+ student community; mentored juniors on system design, Git/GitHub workflows, and production ML engineering.

Projects

Order Matching Engine | C++17, Price-Time Priority, GTC/IOC/FOK, FastAPI, WebSockets | Code, Jun. 2026 – Jul. 2026
• Engineered a dependency-free C++17 limit order book with price-time priority and FIFO per level, supporting LIMIT/MARKET/STOP/STOP-LIMIT and GTC/IOC/FOK semantics with deterministic integer quantities.
• Benchmarked matching throughput at 10.9M orders/sec (0.09µs/order); exposed a trade tape + depth feed over a FastAPI/WebSocket bridge; CI builds & tests on every PR.
SentinelX – AI Threat Detection Platform | FastAPI, React, DistilBERT, Celery, Redis, PostgreSQL | Code, May 2026
• Built a production-grade, multi-channel threat-detection backend (email/SMS/voice) with a weighted 4-signal risk engine and async Celery + Redis task queues.
• Integrated HuggingFace DistilBERT + OpenAI Whisper STT; shipped JWT auth, RBAC, and Docker Compose; 1st Place Hack to Future 3.0 (600+ teams, INR 50,000).
Traceify – Error Tracking & Observability SDK | Python, FastAPI, React, PostgreSQL, CI/CD | Code, Apr. 2026
• Designed and published a multi-language error-tracking SDK (PyPI traceify-sdk) with native Python, Java, Kotlin, Go, C++, JavaScript clients.
• Built a multi-tenant FastAPI backend with Redis caching and LLM-powered error reports; 1st Runner-Up Orchathon 2K26 (800+ teams).

Open Source

- Merged 50+ pull requests into high-profile open source repos including Google (OSS-Fuzz), Microsoft (VibeVoice), Apache (DataFusion), and Hedera Hashgraph (Hiero SDK).
- Hiero SDK (Hedera): Refactored the EVM transaction builder to support direct alias conversion, cutting boilerplate by 30% and eliminating type redundancy errors (PR #1030).

Technical Skills

Languages: Python, Java, C++, JavaScript (ES6+), TypeScript, SQL
Frameworks: FastAPI, Node.js, Express.js, React.js, Next.js, Flask
Infrastructure: Docker, Kubernetes, Redis, Kafka, GitHub Actions, AWS (EC2/S3/Lambda), Linux
Core Concepts: DSA (366+ LeetCode), Distributed Systems, Microservices, CI/CD, Asynchronous Programming, RESTful APIs

Achievements & Certifications

Competitive Programmer: 366+ LeetCode problems solved (104 Easy, 176 Medium, 86 Hard) + 50 SQL problems.
IEEE First-Author Publication presented at ICCTWC 2026; 1st Place at Hack to Future 3.0 (600+ teams) and 1st Runner-Up at Orchathon 2K26 (800+ teams).
Certifications: JPMorgan Chase & Electronic Arts Software Engineering Virtual Experience; NPTEL Programming in Java (Elite+Gold, 91/100).